



RESEARCH REPORT

Vision-Based UAV Self-Positioning in Low-Altitude Urban Environments³

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³Dai et al. [1]

1 Evaluation Metrics

- Ranking Process
- Spatial Discrete Index ($R@K$)
- Spatial Discretization Matrix ($SDM@K$)

Evaluation Metrics

After the training process, the model outputs an optimal representation z_i for each input aerial image i .

Vector Space Model

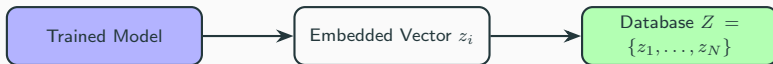
- Documents and queries are represented as vectors in high-dimensional space
- Similarity between them is measured to retrieve the most similar documents
- Each label in our dataset is treated as a document with representation z_i

Embedded Vector Database

$$Z = \{z_1, z_2, \dots, z_N\}$$

Used in the retrieval process with the query (real-time image from UAV).

Pre-ranking Process



Two-Part Ranking Process

- ❶ **Evaluation:** Use queries from query dataset to evaluate model performance
- ❷ **Real-time Retrieval:** Apply ranking for UAV self-positioning task

Definition

$$\text{sim}(l_i, l_j) = \cos(\theta_{l_i, l_j}) = \frac{\vec{z}_i \cdot \vec{z}_j}{|\vec{z}_i| |\vec{z}_j|}$$

- θ_{z_i, z_j} : angle between z_i and z_j
- Dot product: $\sum_{k=1}^d z_i[k] z_j[k]$
- Euclidean length: $|\vec{z}_i| = \sqrt{\sum_{k=1}^d z_i[k]^2}$

Normalized Form

Let $\vec{v}(z_i) = \frac{\vec{z}_i}{|\vec{z}_i|}$, then:

$$\text{sim}(l_i, l_j) = \vec{v}(z_i) \cdot \vec{v}(z_j)$$

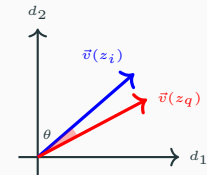
Score Function

For query vector z_q and embedded vector z_i :

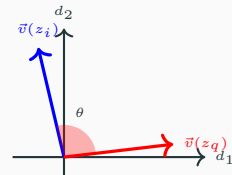
$$\text{score}(l_q, l_i) = \frac{\vec{V}(z_q) \cdot \vec{V}(z_i)}{\|\vec{V}(z_q)\| \|\vec{V}(z_i)\|} = \vec{v}(z_q) \cdot \vec{v}(z_i)$$

- Score function = cosine similarity $\cos(\theta_{l_q, l_i})$
- $i = \overline{1, N}$ where N = number of embedded vectors in database Z
- **Higher similarity score** $\Rightarrow$ more similar vectors

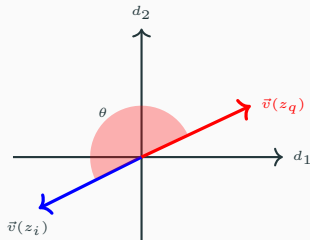
Cosine Similarity - Visual Illustration



Similar ($\cos \theta \approx 1$)



Unrelated ($\cos \theta \approx 0$)



Opposite ($\cos \theta \approx -1$)

Key Insight

When query vector z_q is similar to embedded vector z_i , the query's location is also similar to the embedded vector's location.

Precision vs Recall

	Relevant	Nonrelevant
Retrieved	tp	fp
Not retrieved	fn	tn

Definitions

$$\text{Precision} = \frac{tp}{tp + fp}$$

$$\text{Recall} = \frac{tp}{tp + fn}$$

Focus on Recall

In UAV self-positioning and image retrieval, we are more interested in **Recall** evaluation.

Indicator Function

Given query image with label l_q , for the i -th ranked embedded vector:

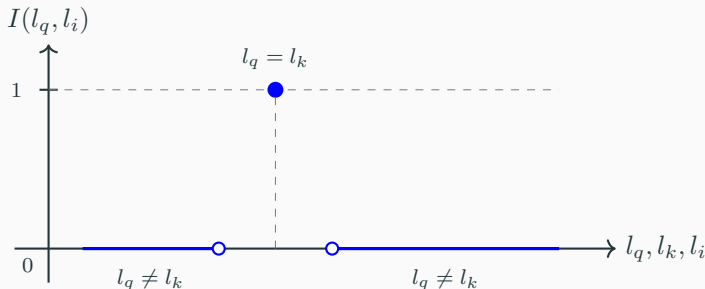
$$I(l_q, l_i) = \begin{cases} 1 & \text{if } \exists k \in \overline{1, i} \text{ such that } l_q = l_k \\ 0 & \text{otherwise} \end{cases}$$

R@K Definition

For a set of query images Q :

$$\text{R@K} = \frac{1}{\|Q\|} \sum_{q \in Q} I(l_q, l_K)$$

Indicator Function - Visual



R@1 Interpretation

$$R@1 = \frac{1}{\|Q\|} \sum_{q \in Q} I(l_q, l_1)$$

Proportion of queries where the top-1 retrieved location matches the query's location.

Main Disadvantage

R@K is a **binary metric** - very rigid with a hard threshold:

- If retrieved location is even slightly incorrect $\Rightarrow$ indicator = 0
- No tolerance for small errors

Solution

Propose SDM@K metric - a **continuous metric** that allows very small errors in the retrieved location.

Modified Indicator Function

Replace discrete indicator with continuous smooth function using e^{-x} :

$$\text{SDM}_k = (K - k + 1) \times e^{-s \times d_k} = \frac{K - k + 1}{e^{s \times d_k}}$$

- $K - k + 1$: weight for higher ranks
- $s = 5 \times 10^3$: scaling factor
- d_k : distance to k -th embedded vector

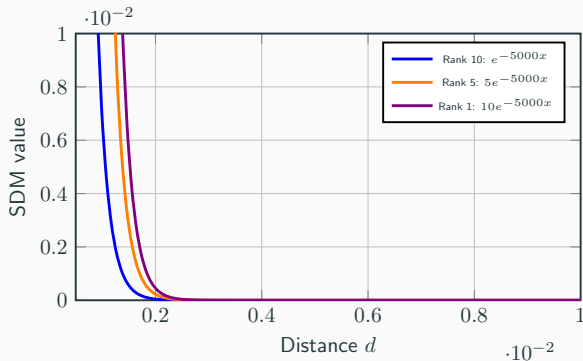
Distance d_k

$$d_k = \sqrt{(x_q - x_k)^2 + (y_q - y_k)^2}$$

(x_q, y_q) : query location

(x_k, y_k) : k -th vector location

SDM Weighting - Visualization



Key Insight

Higher-ranked vectors get more weight and greater acceptable error tolerance.

Final Formula

Normalized to range $[0, 1]$:

$$\text{SDM@K} = \frac{\sum_{i=1}^K \frac{K - i + 1}{e^{s \times d_i}}}{\sum_{i=1}^K (K - i + 1)}$$

Evaluation on Query Set Q

Compute SDM@K for each query image in Q and divide by total number of queries.

Main Advantage

SDM@K is a **continuous metric** - not as rigid as R@K and allows for very small errors in the retrieved location.

Comparison: R@K vs SDM@K

R@K

- Binary metric (0 or 1)
- Hard threshold
- No tolerance for small errors
- Simple interpretation

SDM@K

- Continuous metric
- Smooth decay with distance
- Tolerates small errors
- Weighted by rank

Property	R@K	SDM@K
Type	Discrete	Continuous
Error tolerance	None	Yes
Rank weighting	No	Yes
Range	$\{0, 1\}$	$[0, 1]$

- [1] Ming Dai et al. **“Vision-Based UAV Self-Positioning in Low-Altitude Urban Environments”**. In: *IEEE Transactions on Image Processing* 33 (2024), pp. 493–508.
- [2] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. **Introduction to Information Retrieval**. Cambridge, UK: Cambridge University Press, 2008. ISBN: 978-0-521-86571-5. URL: <http://nlp.stanford.edu/IR-book/information-retrieval-book.html>.
- [3] Liang Zheng et al. **“Scalable Person Re-Identification: A Benchmark”**. In: *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. Dec. 2015.

Thanks for listening!